Technical Datasheet - VBF-T6LC-DSH-01

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Field	Value	Source
Rated capacity (Ah)	4.5 (nominal); 4.804 verified	`data/LNMO.json`; `final_1c_dfn_h45.json:capacity_ah`
Nominal voltage / voltage window (V)	2.5 - 4.7	`data/LNMO.json`
Rated energy (Wh)	19.94	`final_1c_energy_dfn_h45.json:energy_wh`
Volumetric energy density (Wh/L)	1005.8	`final_1c_energy_dfn_h45.json:energy_density_wh_l`
Gravimetric energy density (Wh/kg)	480.7 (electrolyte/casing excluded)	`final_1c_energy_dfn_h45.json:energy_density_wh_kg`
Voltage plateau (discharge midpoint, V)	4.115	`final_1c_energy_dfn_h45.json:midpoint_voltage_v`
Maximum continuous discharge rate	1C (verified 4.804 Ah)	1C_discharge simulation
Fast-charge capability	4C at 45 degC : T_max 322.59 K, no lithium plating (anode potential min +0.0894 V)	`final_4c_safety_h45_dfn.json`
Cycle life (SEI)	100 cycles: SEI 9.5 nm (DFN) / 233 nm (SPMe); <= 500 nm	`final_aging_dfn_h45.json`; `R5A_aging_k1e-15.json`
Operating temperature range	simulated at 25 degC (discharge), 45 degC (4C charge); broader range N/A	simulation protocols
Safety determination	no plating at 4C; T_max 322.59 K <= 50 degC	`final_4c_safety_h45_dfn.json`
Dimensions	electrode strip 65 mm x 1580 mm x 0.193 mm (unwound); pouch shell Not provided	parameter set
Mass	41.48 g (excl. electrolyte/casing); +6.2 g electrolyte (literature density)	`final_1c_energy_dfn_h45.json:mass_kg`
DC resistance (10% discharge)	7.0 mΩ	`final_1c_energy_dfn_h45.json:dcr_ohm`
Chemistry	LNMO spinel / graphite	`data/LNMO.json`